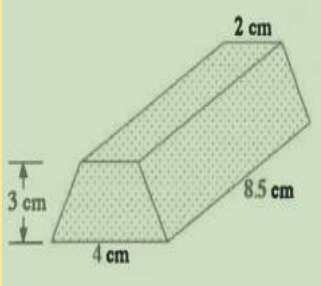
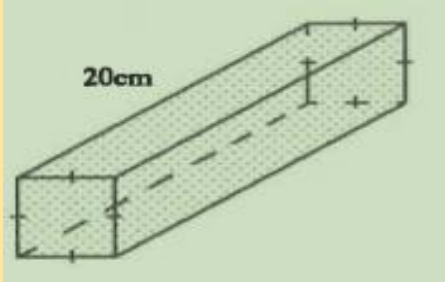
WORK THE FOLLOWING QUESTION TO UNDERSTAND MORE

1. The volume of the solid figure given below is equal to



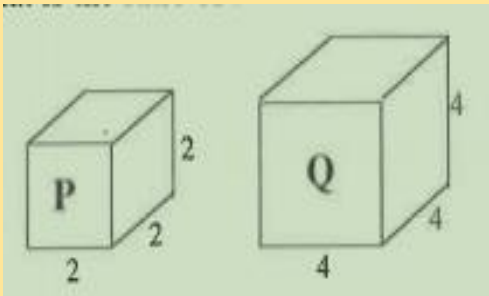
A. 76.5
B. 128.5
C. 64. 5
D. 45

2. What is the total surface area, if the volume of the diagram given below is $500cm^3$.



A. $25cm^2$
B. $400cm^2$
C. $4500cm^2$
D. $50cm^3$

3. The ratio of the volume of cube P to cube Q?



A. 1: 8
B. 1: 4
C. 1: 2
D. 2: 1

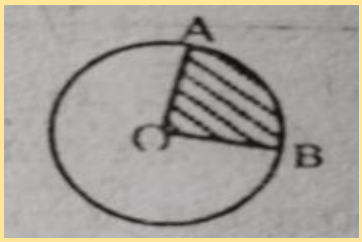
ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE

2005 Mathematics Examination

4. A right circular cylinder with radius of the base 5cm and height 8cm is open from the top. What is the total surface area of this cylinder?

A. $105\pi cm^2$ B. $200\pi cm^2$ C. $130\pi cm^2$ D. $80\pi cm^2$

5. In the circle shown below: O is the center,the radius measures 9cm and $m(\angle AOB) = 70^\circ$. Which one of the following is the area of the unshaded region?



A. $\frac{261}{4}\pi cm^2$ B. $\frac{9}{40}\pi cm^2$ C. $\frac{64}{4}\pi cm^2$ D. $\frac{63}{4}\pi cm^2$

6. The height of a right circular cylinder is 6cm. If the radius of its base is 3cm. what is the total surface area of this cylinder?

A. $45\pi cm^2$ B. $54\pi cm^2$ C. $64\pi cm^2$ D. $36\pi cm^2$

7. The base of a regular pyramid is an equilateral triangle of sides 6 units long. If the altitude of this pyramid is 12 units, then find its volume.

A. $36\sqrt{3}$ cubic unit C. $192\sqrt{3}$ cubic unit
B. $128\sqrt{3}$ cubic unit D. $64\sqrt{3}$ cubic unit

8. Find the total surface area of a hemisphere with radius 6 units long.

A. 36π squre unit C. 72 square unit
B. 108 square unit D. 144 square unit

9. If a frustum is formed from a regular pyramid, then which one of the following statements is true about its lateral faces?

A. They are isosceles trapezium C. They are rhombus
B. Their altitudes are not equal D. Their areas are not equal

10. The lower and upper bases of a frustum of a regular pyramid are squares with measures of sides 4 meters and 3 meters respectively. If the altitude of the frustum is 6 meters,then its volume is equal to _____

A. $\frac{54+4\sqrt{3}}{3}m^3$ C. $\frac{37+\sqrt{37}}{3}$ m^3
B. $\frac{37\sqrt{37}}{3}$ m^3 D. 60 m^3

ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE

2006 Mathematics Examination

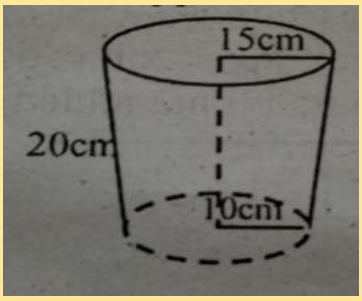
11. Depending on question #7 the total surface area is equal to

A. $(7\sqrt{145}+25)cm^2$ C. $(14\sqrt{145}+50)cm^2$
B. $(28\sqrt{145}+25)cm^2$ D. $(4\sqrt{145}+25)cm^2$

12. If the diameter of a sphere is 16 meters, then its surface area is equal to :

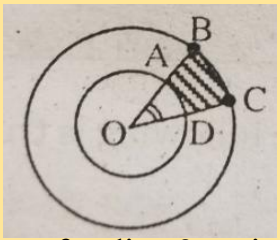
A. $256\pi m^2$ B. $64\pi m^2$ C. $1024\pi m^2$ D. $128\pi m^2$

13. The figure shown below is a container made from an inverted frustum of a right circular cone. The radius of its lower base is 10cm and that of the upper base is 15 cm. If the height of this container is 20 cm, then which one of the following is equal to its volume?



A. $5000\pi cm^3$ B. $7500\pi cm^3$ C. $\frac{9500}{3}\pi cm^3$ D. $\frac{2500}{3}\pi cm^3$

14. Suppose two concentric circles of radii of 3cm and 6cm with a shaded region are drawn as shown in the figure below. If m(∠BOC) = 40˚, then what is the area of the shaded region?



A. $3\pi cm^2$
B. πcm^2
C. $6\pi cm^2$
D. $4\pi cm^2$

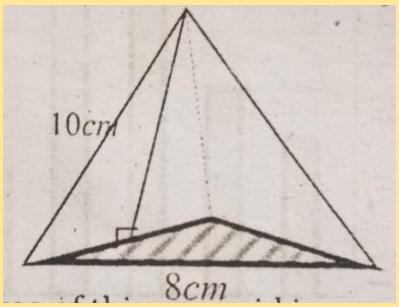
15. A wooden sphere of radius 3cm is drilled along its diameter by a metal driller. The driller has a shape of right circular cylinder of radius 1cm. after the drilling, what is the volume of the remaining part of the sphere?

A. $30\pi cm^3$ B. $42\pi cm^3$ C. $24\pi cm^3$ D. $35\pi cm^3$

ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE

2007 Mathematics Examination

16. The figure shown below is a right regular pyramid. The length of its slant height is 10cm and its base is an equilateral triangle with side 8cm long. The total surface area of this pyramid is equal to_____



A. $(12+\sqrt{3})cm^2$
B. $(120+3\sqrt{3})cm^3$
C. $(120+\sqrt{3})cm^2$
D. $(120+16\sqrt{3})cm^2$

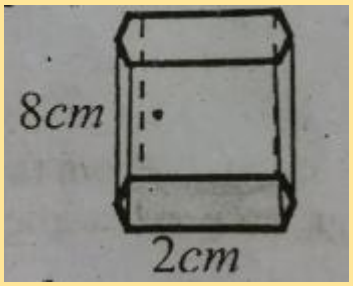
17. The following four statements are about frustum of a regular pyramid.

I. The lateral faces of a frustum of a regular pyramid are isosceles trapezium.
II. The slant height of the frustum of a regular pyramid is the edge of the lateral faces.
III. The lateral faces of a frustum of a regular pyramid are triangles.
IV. The slant height of the frustum of a regular pyramid is the altitude of any one of the lateral faces.

Which group of the above statements correctly describes properties of the frustum of a regular hexagonal pyramid?

A. I and IV only C. II and IV only
B. I, II, III and IV **D**. I, II and IV only

18. As shown in the figure below the height of a right prism is 8cm and its base is a regular hexagon of side 2cm. What is the volume of this prism?



A. $15\sqrt{3}cm^3$
B. $60cm^3$
C. $48\sqrt{3}cm^3$
D. $30cm^3$

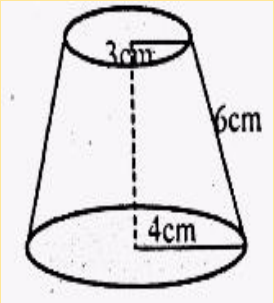
ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE

2008 Mathematics Examination

19. A right circular cylinder, which is closed at its upper and lower base, has height of 8cm. if the diameter of the bases is 8cm, then what is its total surface area?

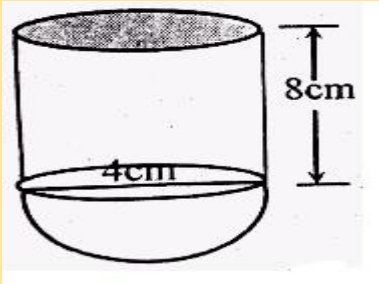
A. $256\pi cm^2$ B. $122\pi cm^2$ C. $96\pi cm^2$ D. $48\pi cm^2$

20. Frustum of a right circular cone with base radii 3cm and 4cm and a slant height of 6cm given as shown below. What is the total surface area of this frustum?



A. $67\pi cm^2$
B. $74\pi cm^2$
C. $56\pi cm^2$
D. $36\pi cm^2$

21. If a right circular cylinder of height 8cm is placed on top of a hemispherical iron ball of radius 2cm as shown on the figure below. What is the volume of this solid?

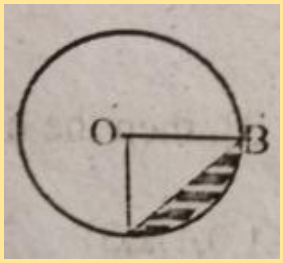


A. $\frac{840}{3}\pi cm^3$
B. $\frac{112}{3}\pi cm^3$
C. $\frac{256}{3}\pi m^3$
D. $\frac{128}{3}\pi cm^3$

22. The volume of a right circular cone of height 12cm and a slant height 13cm is equal to:
A. $3000\pi cm^3$ B. $200\pi cm^3$ D. $75\pi cm^3$ D. $100\pi cm^3$

23. Find the volume of right square prism whose length of sides of the base is 6cm and height 10cm.
A. $312cm^3$ B. $360cm^3$ C. $430cm^3$ D. $240cm^3$

24. In the figure given below, AB is a chord of a circle with center O and radius 6cm. If $m(\angle AOB) = 90^\circ$, then what is the perimeter of the shaded region?

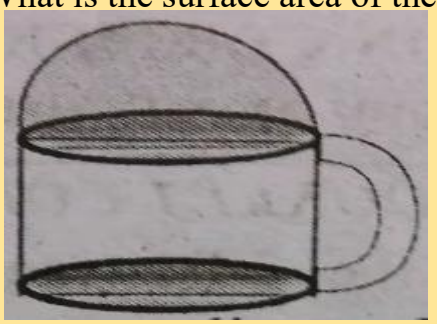


A. $(3\pi + 6\sqrt{2})cm$
B. $(9\pi + 12)cm$
C. $(9\pi + 6\sqrt{6})cm$
D. $(3\pi + 12)cm$

ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE

2009 Mathematics Examination

25. A cylindrical teacup of base radius 4cm and height 10cm has a hemispherical lid (top cover).What is the surface area of the lid part?



A. $16\pi cm^2$
B. $32\pi cm^2$
C. $40\pi cm^2$
D. $64\pi cm^2$

26. A regular pyramid has a square base whose side is 8 cm long. If the length of the lateral edge is 12 cm, then which one of the following is the total surface area?

A. $64(1+2\sqrt{2})cm^2$ C. $64\sqrt{5}\text{cm}^2$

B. $(64+128\sqrt{2})\text{cm}^2$ D. $64(1+\sqrt{5})cm^2$

27. A frustum of height 12 cm is formed from right circular cone of height 16 cm and base radius 8 cm. the volume of the frustum is:

A. $330\pi\text{cm}^3$ C. $336\pi\text{cm}^3$

B. $104\pi\text{cm}^3$ D. $312\pi\text{cm}^3$

28. A right circular cylinder with base diameter 10 cm contains enough water that can immerse an object. If an iron ball of diameter 6 cm is immersed, how high will the water rise?

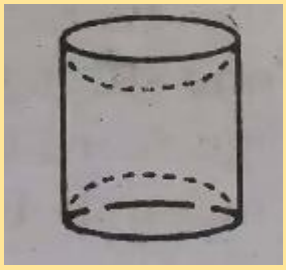
A. $\frac{12}{25}$cm C. $\frac{48}{25}$cm

B. $\frac{36}{25}$cm D. $\frac{72}{25}$cm

ETHIOPIAN GENERAL SECONDARY EDUCATION CERTIFICATE

2010 Mathematics Examination

29. A solid is formed by a cylinder with radius 3 cm and height 5 cm and by two hemispheres as shown below. Which of the following is the volume of the solid object?



A. $16\pi\text{cm}^3$ C. $12\pi\text{cm}^3$

B. $15\pi\text{cm}^3$ D. $9\pi\text{m}^3$

30. A right circular cone of base radius 4 cm and height 10 cm is cut off by a plane parallel to the base at a distance of 5 cm from the base. What is the volume of the resulting frustum?

A. $\frac{140}{3}\pi cm^3$ C. $\frac{160}{3}\pi cm^3$

B. $\frac{28}{3}\pi cm^3$ D. $16\sqrt{21}\pi\text{cm}^3$

31. On question **#26** the total surface area is equal to

A. $(6\sqrt{26}+20)\pi cm^3$ C. $3(2\sqrt{29}+10)\pi\text{cm}^2$

B. $2(3\sqrt{29}+10)\pi\text{cm}^2$ D. $(6\sqrt{29}+20)\pi\text{cm}^2$

32. Which of the following is the volume of a sphere whose surface area is $144\pi\text{cm}^2$?

A. $288\pi\text{cm}^3$ C. $162\pi\text{cm}^3$

B. $268\pi\text{cm}^3$ D. $122\pi\text{cm}^3$

33. The base of a right triangular prism is an isosceles triangle of sides 7cm, 6cm, and 7cm long. If the height of the prism is 9cm, what is the volume of this prism?

A. $54\sqrt{10}\text{cm}^3$ C. $27\sqrt{10}\text{cm}^3$

B. $18\sqrt{10}\text{cm}^3$ D. $21\sqrt{10}\text{cm}^3$